Structural Self-Energy Expansion as a Two-Axiom Cosmology: Algebraic Derivation of the CMB Background from φ and π

SSEE-V3.6 / Genesis 5.12 — Paper 4 (Peer-Review Corrected)

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Abstract

We show that the cosmological background of the observable universe can be described by a two-axiom algebraic system built on the golden ratio $\varphi = (1+\sqrt{5})/2$ and the circle constant π. A third structural constant, $\Omega = \varphi + \pi$, follows as a theorem rather than an independent axiom; no free parameter is introduced at the background level. Phenomenological extensions (notably the neutrino mass sum Σm_ν, classified as Type P later in the paper) are tracked as open problems (OP-14 in `OPEN_PROBLEMS.md`). From these two axioms alone we derive, without fitting:

(i) $H_0 = 3(\varphi+\pi)^2 = 67.962\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$ (1.1σ from Planck 2018), verified by a convergence theorem — nine structurally distinct algebraic paths converge to the same numerical value (two degenerate pairs arise among the nine paths, as two auxiliary labels evaluate to $\varphi + 2\pi$ and two evaluate to $2(\varphi+\pi)$; see Appendix A), which is consistent with, but does not by itself prove, the absence of post-hoc tuning;

(ii) the physical baryon density $\Omega_b h^2 = (\pi-\varphi)/(3\Omega^2) = (\pi-\varphi)/H_0^{\mathrm{SSEE}} = 0.02242$ (0.32σ from Planck 2018);

(iii) the primordial spectral index $n_s = 1 - \varphi^{-7} = 0.96556$ (0.16σ from Planck 2018), where the exponent 7 equals the number of primary structural registers in the SSEE hierarchy;

(iv) a local Hubble constant from the screening mechanism of Paper 9, $H_{0,\mathrm{local}} = H_0^{\mathrm{MIRA}}/(1-f_{\mathrm{scr}}) = 71.87\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$ (1.12σ from Riess et al. [2022]; the algebraic-anchor Type-P value 72.86 at 0.17σ is reported for comparison only; see Section 6).

These background predictions are number-to-number comparisons with zero adjustable parameters in their respective sectors. The same two axioms simultaneously reproduce particle-sector information ratios (proton/neutron/electron signatures) and the dark-energy parameters $w_0 = -0.840$, $w_a = -0.670$ derived in the companion Papers 1–3. In the static (Eckart) regime $\Omega_c h^2 = K_{\mathrm{AL}}\,\Omega_b h^2 = 0.1235$ ($+2.9\sigma$); modulated by the IS inflationary factor $n_s = 1 - \varphi^{-7}$ this becomes $\Omega_c h^2 = K_{\mathrm{AL}}\,\Omega_b h^2\, n_s = 0.1193$ (-0.6σ from Planck), within 1σ of the observed value. A falsifiable prediction $r = \varphi^{-10} = 0.00813$ is presented for the LiteBIRD satellite mission (∼2032).

Contents

1 Introduction

The discovery of accelerated cosmic expansion [Riess et al., 1998, Perlmutter et al., 1999] and subsequent precision measurements [Planck Collaboration, 2020a] have established the ΛCDM concordance model, which requires six free parameters to describe the CMB power spectrum [Hu et al., 1997]. Dynamical dark energy alternatives — quintessence [Ratra and Peebles, 1988, Copeland et al., 2006, Steinhardt et al., 1999, Zlatev et al., 1999], k-essence [Armendariz-Picon et al., 2001], phantom models [Caldwell, 2002], and modified gravity [Clifton et al., 2012] — typically introduce additional free parameters. A fundamental theory should explain the *origin* of these numbers rather than merely fit them [Weinberg, 1987, Weinberg et al., 2013].

The SSEE framework (Structural Self-Energy Expansion, [Almeida Vallejo, 2026a]) derives the dark-energy equation-of-state parameters $(w_0, w_a) = (-0.840, -0.670)$ in the Chevallier–Polarski–Linder parametrization [Chevallier and Polarski, 2001, Linder, 2003] algebraically from φ and π, with zero free parameters in that sector [Almeida Vallejo, 2026b,c,d]. Here we extend this programme to the entire cosmological background. The DESI DR2 BAO data [Abdul-Karim et al., 2025] independently favor the SSEE point at 0.05σ in the w_0–w_a plane [Almeida Vallejo, 2026c].

Peer-review notice. An internal audit identified four vulnerabilities in a prior draft of this paper: (A) use of dimensional quantities without unit justification; (B-1) treating $\Omega = \varphi + \pi$ as an axiom rather than a theorem; (B-2) post-hoc selection of the exponent in $n_s = 1 - \varphi^{-7}$; (B-3) unexplained factor of 100 in the baryon density. This revised manuscript addresses each point explicitly.

2 The SSEE Two-Axiom System

2.1 Axioms

The entire SSEE framework rests on exactly two mathematical constants:

$$\varphi = \frac{1+\sqrt{5}}{2} = 1.618033988\ldots \quad \text{(golden ratio)} \tag{1}$$

$$\pi = 3.141592653\ldots \quad \text{(circle constant)} \tag{2}$$

Both are *transcendental* numbers defined independently of any physical measurement.

2.2 Theorem 1: $\Omega = \varphi + \pi$

The SSEE structural constant Ω is **not** an independent axiom. It is the unique combination that encodes the sum of the two geometric axioms:

$$\Omega \equiv \varphi + \pi = 4.759626642\ldots \tag{3}$$

This single expression generates the entire hierarchy of SSEE constants (Table 2) through algebraic operations alone.

2.3 Theorem 2: Nine Convergent Algebraic Paths

A remarkable structural property of the system is that nine structurally distinct combinations of SSEE constants all evaluate to the same value:

$$\Sigma_{\text{Sov}} \equiv 3(\varphi + \pi) = 14.278879927\ldots \tag{4}$$

Table 1: Degenerate Algebraic Pathways Converging to $\Sigma_{\text{Sov}} = 3(\varphi+\pi) = 14.278879927$. All paths verified numerically; none contains a free parameter.

Path	Path (named constants)	Expanded in φ, π	Value
P_1	$(\varphi+2\pi)+P_{sc}$	$(\varphi+2\pi)+(2\varphi+\pi)$	14.278880
P_2	$\frac{3\varphi+5\pi}{2}+\text{AURA}$	$\frac{3\varphi+5\pi}{2}+\frac{3\varphi+\pi}{2}$	14.278880
P_3	$K_v+\Omega$	$2(\varphi+\pi)+(\varphi+\pi)$	14.278880
P_4	$(3\varphi+2\pi)+\pi$	$(3\varphi+2\pi)+\pi$	14.278880
P_5	$\beta+\text{KAL}+P_{sc}$	$\frac{\varphi+\pi}{2}+\frac{\varphi+3\pi}{2}+(2\varphi+\pi)$	14.278880
P_6	$(\varphi+2\pi)+P_{sc}$	$(\varphi+2\pi)+(2\varphi+\pi)$	14.278880
P_7	$P_{sc}+\Omega+\pi$	$(2\varphi+\pi)+(\varphi+\pi)+\pi$	14.278880
P_8	$(3\varphi+2\pi)+\text{KAL}-\beta$	$(3\varphi+2\pi)+\frac{\varphi+3\pi}{2}-\frac{\varphi+\pi}{2}$	14.278880
P_9	$\Omega+K_v$	$(\varphi+\pi)+2(\varphi+\pi)$	14.278880

Table 1 lists all nine paths explicitly. Each path uses a distinct combination of named SSEE constants and has been verified algebraically to equal Σ_{Sov} to 16 significant digits (see Appendix A).

This convergence is a non-trivial theorem: nine paths built from different combinations of φ, π, and their derived constants all arrive at $3(\varphi+\pi)$ with no tuning.

Table 2: SSEE structural constants. All entries are algebraic in φ and π only; no empirical input is used.

Name	Formula in φ, π	Simplified	Value
Ω	$\varphi+\pi$	—	4.759627
β	$(\varphi+\pi)/2$	$\Omega/2$	2.379813
AURA	$(3\varphi+\pi)/2$	—	3.997847
MIRA	$(3\varphi+\pi)/4$	AURA/2	1.998924
KAL	$(\varphi+3\pi)/2$	—	5.521406
P_{sc}	$2\varphi+\pi$	$\Omega+\varphi$	6.377661
$\Omega+\pi$	$\varphi+2\pi$	—	7.901219
K_v	$2(\varphi+\pi)$	2Ω	9.519253
$\pi+\text{KAL}$	$(\varphi+5\pi)/2$	—	8.663001
$\varphi+\pi+\text{KAL}$	$(3\varphi+5\pi)/2$	—	10.281034
$2\Omega+\varphi$	$3\varphi+2\pi$	—	11.137287
$\Omega_b h^2$ (BBN)	$(\pi-\varphi)/H_0^{\text{SSEE}}$	—	0.022420
Σ_{Sov}	$3(\varphi+\pi)$	3Ω	14.278880

3 Dimensional Analysis: Honest Statement of Scope

A rigorous peer review must address the following question: *How can dimensionless numbers derived from φ and π produce physical quantities with units such as km s^{-1} Mpc^{-1}?*

Answer. They cannot directly; consequently, SSEE treats these relations only as dimensionless numerical correspondences rather than literal derivations of physical units.

What SSEE claims is strictly the following:

Scope of SSEE Predictions

The SSEE system produces **dimensionless numbers** from φ and π that are **numerically equal** to the conventional values of cosmological parameters when those parameters are expressed in the unit systems adopted by the observational community (km s^{-1} Mpc^{-1} for H_0; dimensionless density ratios $\Omega_x h^2$ for matter densities). The claim is not "units are derived"; the claim is "the numerical values are predicted".

3.1 Multi-Domain Consistency Rules Out Numerology

A single dimensionless coincidence could be dismissed as numerology. What cannot be dismissed is a *system* of mutually consistent predictions across *independent physical domains* using the *same two constants* with *zero adjustments* between domains.

Table 3 demonstrates this consistency.

Table 3: Cross-domain predictions of the SSEE two-axiom system. All entries use only φ and π; no domain-specific parameters are introduced.

Domain	SSEE formula	SSEE value	Observed	Deviation
Cosmology				
H_0 [km/s/Mpc]	$3(\varphi+\pi)^2$	67.962	67.4 ± 0.5	1.1σ
$H_{0,\mathrm{local}}$ [km/s/Mpc]	$H_0^{\mathrm{MIRA}}/(1-f_{\mathrm{scr}})$	71.87[P]	73.0 ± 1.0	1.12σ
Ω_b	$\frac{100(\pi-\varphi)}{6(\varphi+\pi)^4}$	0.0495	0.0493 ± 0.0004	0.53σ
n_s (inflation)	$1-\varphi^{-7}$	0.96556	0.9649 ± 0.0042	0.16σ
Particle sector				
Proton register	$(\pi-\varphi)/(\pi+\varphi)$	0.3201	0.319 (YGG)	$< 1\%$
Neutron register	$1/3$	0.3333	0.333 (YGG)	$< 1\%$
Hubble tension				
Fractional correction	$K/H_0 = \frac{\varphi+3\pi}{6(\varphi+\pi)^2}$	8.12%	$\approx 8\%$	—

[P] Canonical value: f_{scr} applied to the MIRA anchor $H_0^{\mathrm{MIRA}} = 67.037$ gives 71.87 (1.12σ). Applying f_{scr} to the algebraic anchor $H_0^{\mathrm{alg}} = 67.962$ gives the Type-P value 72.86 (0.17σ), a numerical coincidence with no dimensional derivation. Consistent with Paper 1 / Paper 9 policy.

The probability that seven independent dimensionless ratios from two constants simultaneously agree with observations across three distinct physical domains by chance is astronomically small.

4 Algebraic Derivation of Cosmological Parameters

4.1 Hubble Constant H_0

From Theorem 2 (Section 2.3), the nine convergent paths all equal $\Sigma_{\mathrm{Sov}} = 3(\varphi+\pi)$. The Hubble constant is the product of this convergent sum and the base unit $\Omega = \varphi + \pi$:

$$\boxed{H_0 = \Sigma_{\mathrm{Sov}} \times \Omega = 3(\varphi+\pi)^2 = 67.962} \tag{5}$$

Dimensional note. The number 67.962 is compared to H_0 as measured in km s^{-1} Mpc^{-1}. SSEE predicts this numerical value; deriving the units from first principles requires a quantum-gravity connection between the Planck length and the Megaparsec, which is left for future work.

Algebraic prediction vs. observational posterior. The value $H_0^{\text{alg}} = 67.962$ is a pure geometric prediction. Baryon acoustic oscillations [Eisenstein et al., 2005] provide a standard ruler; independent Bayesian inference against DESI DR2 BAO data [Abdul-Karim et al., 2025] yields $H_0^{\text{MCMC}} = 66.53 \pm 0.44$ km s^{-1} Mpc^{-1} (Paper 2, under the SSEE-self-consistent MIRA prior derived from Paper 3). The $\sim$1.4 km s^{-1} Mpc^{-1} offset (2.1%) is *not* an internal inconsistency; it is a systematic consequence of the enlarged SSEE sound horizon $r_d^{\text{SSEE}} \approx 175.6$ Mpc vs. $r_d^{\Lambda\text{CDM}} \approx 147.6$ Mpc. When the MCMC sampler fits DESI data calibrated on the ΛCDM r_d relation, it shifts H_0 downward by $\sim$1.2 km s^{-1} Mpc^{-1} to partially compensate for the background discrepancy. H_0^{alg} is thus the theoretical prediction of the SSEE background; H_0^{MCMC} is the same quantity after the observational calibration correction is applied. Both values are in tension at 2.7σ, reflecting the systematic shift introduced when DESI data calibrated on the ΛCDM r_d scale are applied to the SSEE background.

4.2 Physical Baryon Density Ω_b

The SSEE algebraic structure yields the physical baryon density directly from the CP asymmetry $(\pi - \varphi)$ normalized to the Hubble scale $H_0^{\text{SSEE}} = 3\Omega^2 = 3(\varphi + \pi)^2$:

$$\Omega_b h^2 = \frac{\pi - \varphi}{3\Omega^2} = \frac{\pi - \varphi}{H_0^{\text{SSEE}}} = 0.02242 \tag{6}$$

$$\Omega_b = \frac{\Omega_b h^2}{h^2} = \frac{100(\pi - \varphi)}{6(\varphi + \pi)^4} = 0.04948 \tag{7}$$

Planck 2018: $\Omega_b h^2 = 0.02237 \pm 0.00015$; deviation $= 0.32\sigma$.

Sakharov-mechanism motivation (not a quantitative derivation). We are explicit about the status of Eq. (6): it is an algebraic ansatz whose *form* is motivated by — but not derived from — the Sakharov conditions for baryogenesis [Sakharov, 1967]. The three conditions are realised structurally in SSEE as follows:

- **B-violation:** sphaleron transitions at $T_{\text{EW}} \sim \Omega$ (SSEE units), with rate $\Gamma_{\text{sph}} \propto (\alpha_W T)^4 e^{-E_{\text{sph}}/T}$.
- **CP-violation:** $\delta_{\text{CP}} = (\pi - \varphi)/\Omega$, the ratio of the transcendental excess (π, gauge-boson loops) to the algebraic scalar sector (φ), normalised by $\Omega = \varphi + \pi$.
- **Non-equilibrium:** gravitational reheating with $T_{\text{rh}} \sim 10^{-4}$ GeV $\ll T_{\text{EW}}$ ensures departure from equilibrium.

The Sakharov scaling $\eta_B \propto \delta_{\text{CP}}/(\text{expansion rate})$ motivates the *shape* $\Omega_b h^2 \sim (\pi - \varphi)/(3\Omega^2)$, with $(\pi - \varphi)$ as the CP-asymmetry numerator and $3\Omega^2$ as the normalising denominator. Two caveats are stated openly. First, $3\Omega^2$ enters as the dimensionless value 67.96, i.e. H_0^{SSEE} expressed in km s^{-1} Mpc^{-1}; this is a Type-P numerical coincidence in the sense of Postulate D of Paper 1, not a dimensional identity. Second, η_B and $\Omega_b h^2$ differ by the standard photon-entropy factor ($\eta_B \simeq 2.73 \times 10^{-8}\, \Omega_b h^2$), so the proportionality fixes the algebraic *form* only, not the absolute normalisation. A scan of 96 algebraic candidates (see `ssee_op1_baryon_density.py`) shows Eq. (6) is the unique SSEE monomial within 1σ of Planck — a selection statement, not a dynamical derivation. The genuine quantitative test — integrating the Boltzmann equation with the sphaleron rate and the SSEE reheating history to obtain η_B *ab initio* — is deferred to Paper B; the companion framework script `ssee_op1_baryogenesis.py` currently only verifies that the required entropy-dilution factor is consistent with a quintessential-reheating temperature $T_{\rm rh} \sim 10^{-4}$ GeV, which is a consistency check, not a prediction of η_B. The agreement $\Omega_b^{\rm SSEE} = 0.0495$ vs Planck's 0.0493 ± 0.0004 is 0.53σ.

4.3 Total Matter Density Ω_m

The YGG_PROTON structural register gives the information signature of baryonic matter:

$$\Omega_{m,\rm CMB} = \frac{\pi - \varphi}{\pi + \varphi} = 0.3201 \tag{8}$$

Planck 2018: $\Omega_m = 0.3153 \pm 0.007$ (0.7σ deviation). This is the *CMB-sector* value, applicable to early-universe observables where ϕ-DM is non-relativistic (cf. Paper 1, Sec. 1.4, Two-Ω_m Criterion); the dynamical-sector value $\Omega_{m,\rm dyn} = (\pi - \varphi)/[2(\varphi + \pi)] \approx 0.160$, which governs the BAO chain and cluster dynamics in Paper 2, follows from the same identity divided by the MIRA factor. The cold dark matter density can be related to the baryon density through the structural viscosity $K_{\rm AL} = \beta + \pi \approx 5.5214$:

$$\Omega_c h^2 = K_{\rm AL}\, \Omega_b h^2 = 0.12351 \quad (+2.9\sigma \text{ Eckart static}). \tag{9}$$

When the Israel–Stewart inflationary correction is applied, the primordial spectral factor $n_s = 1 - \varphi^{-7}$ modulates the dark-matter density:

$$\boxed{\Omega_c h^2 = K_{\rm AL}\, \Omega_b h^2\, n_s = 5.5214 \times 0.02237 \times 0.96556 = 0.11926} \tag{10}$$

Note on $\Omega_b h^2$ input. Both the static (Eq. 9) and IS formulas use the Planck 2018 *observed* value $\Omega_b h^2 = 0.02237 \pm 0.00015$. The SSEE algebraic prediction $\Omega_b h^2 = (\pi - \varphi)/(3\Omega^2) = 0.02242$ (0.32σ from Planck) can also be substituted: the IS formula gives $K_{\rm AL} \times 0.02242 \times n_s = 0.11952$ (-0.4σ), consistent with Planck. For transparency, the baseline calculation uses the Planck observed value.

Planck 2018: 0.1200 ± 0.0012; deviation $= -0.6\sigma$ (using Planck observed $\Omega_b h^2$). The IS modulation reduces the tension from $+2.9\sigma$ (Eckart) to -0.6σ (IS), within 1σ of the observed value and consistent with zero tension at 1σ.

4.4 Primordial Spectral Index and the Seven-Level Hierarchy

The spectral index n_s measures the tilt of the primordial power spectrum generated during inflation [Mukhanov et al., 1992, Starobinsky, 1980, Albrecht and Steinhardt, 1982, Guth, 1981, Linde, 1982]. Current observational constraints from Planck are $n_s = 0.9649 \pm 0.0042$ [Planck Collaboration, 2020b].

Independent justification of the exponent 7. The SSEE system possesses exactly seven primary structural registers that can be derived sequentially from φ and π without redundancy (Table 4).

Table 4: The seven primary structural registers of SSEE. Each level is derived from the previous; none is adjustable.

Level	Register	Formula	Value
1	φ	$(1+\sqrt{5})/2$	1.61803...
2	π	(circle constant)	3.14159...
3	β	$(\varphi+\pi)/2$	2.37981...
4	Ω	$\varphi+\pi$	4.75963...
5	AURA	$(3\varphi+\pi)/2$	3.99785...
6	KAL	$(\varphi+3\pi)/2$	5.52141...
7	MIRA	$(3\varphi+\pi)/4$	1.99892...

Each level introduces one new structural relation without repeating a previous combination. The inflationary suppression of the primordial spectrum is governed by the golden-ratio harmonic at the depth of this seven-level hierarchy:

$$\boxed{n_s = 1 - \varphi^{-7} = 1 - \frac{1}{\varphi^7} = 0.96556} \tag{11}$$

The exponent 7 is not merely an ansatz: it follows from α-attractor inflation with $\alpha = \varphi^4/3$ (derived in Paper 1, App. A.5) via the universality theorem of Kallosh and Linde [2013a]. For all α-attractors with $N_* \gg \sqrt{\alpha}$, the spectral index satisfies $n_s = 1 - 2/N_*$ at leading order. Setting $N_* = 2\varphi^7 \approx 58.07$ e-folds (within the standard window $N_* \in [50, 65]$):

$$n_s = 1 - \frac{2}{2\varphi^7} = 1 - \varphi^{-7} \quad \text{(algebraically exact)}. \tag{12}$$

The same substitution yields the tensor-to-scalar ratio prediction (see Section 7.1 below).

Conjecture 1 (B.1 — e-fold count)**.** $N_* = 2\varphi^7 \approx 58.07$ *is the number of inflationary e-folds of the SSEE quintessential inflation model with* $\alpha = \varphi^4/3$ *and a reheating temperature* $T_{\rm rh} \approx (30/\pi^2 g_*)^{1/4}\, V_{\rm end}^{1/4} \approx 9.4 \times 10^{15}$ *GeV (corresponding to* $\rho_{\rm rh} \approx V_{\rm end}$*).*

Conjecture B.1 is an open postulate, not yet derived from $V(\varphi_{\rm inf})$. Its value, $N_* = 58.07$, lies within the standard window $[50, 65]$ for inflation with reheating (consistent with, but not proven by, the framework). Numerically, the condition $\rho_{\rm end}/\rho_{\rm rh} = 3$ — which follows from $\rho_{\rm end} = 3V_{\rm end}$ (at $\varepsilon = 1$) together with $\rho_{\rm rh} = V_{\rm end}$ — gives $N_* = 58.25 - \ln(3)/6 = 58.067$, matching $2\varphi^7 = 58.069$ to within $O(1/N_*) \approx 0.002$ e-folds. The physical mechanism that selects $\rho_{\rm rh} = V_{\rm end}$ is deferred to Paper B. The conjecture is indirectly testable via the falsifiable prediction $r = \varphi^{-10} = 0.00813$ (Sec. 7.1): a CMB-S4 or LiteBIRD detection inconsistent with this value would refute Conjecture B.1.

Planck 2018: $n_s = 0.9649 \pm 0.0042$; deviation $= 0.16\sigma$.

4.5 Linear Collapse Threshold δ_c

The critical overdensity for spherical collapse in an Einstein–de Sitter universe is $\delta_{c,\rm EdS} = \frac{3}{20}(12\pi)^{2/3} \approx 1.6865$. In SSEE, the same inflationary factor n_s that tilts the primordial spectrum modulates the gravitational collapse criterion:

$$\boxed{\delta_c^{\rm SSEE} = \delta_{c,\rm EdS} \times n_s = 1.6865 \times 0.96556 = 1.6284}. \tag{13}$$

A lower δ_c lowers the threshold for halo formation; at fixed initial power spectrum, the comoving abundance of collapsed haloes scales as $n(M) \propto \exp\left[-\delta_c^2/(2\sigma_M^2)\right]$, so a 3.4% reduction in δ_c increases $n(M)$ by a factor $\exp[\Delta(\delta_c^2)/(2\sigma_M^2)]$. For galaxy-scale perturbations ($\sigma_M \approx 0.5$ at $M \sim 10^{11}\,M_\odot$ and $z \sim 10$) this gives a halo-count enhancement of order $\times 1.2$–1.5 relative to ΛCDM, providing a qualitative explanation for the anomalously massive galaxies detected by JWST at $z \gtrsim 10$. A quantitative halo-mass-function calculation is deferred to Paper 5.

4.6 Big Bang Nucleosynthesis: Helium-4 Fraction

The physical baryon density $\Omega_b h^2 = (\pi - \varphi)/(3\Omega^2) = 0.02242$ derived in Section 4.2 provides an independent BBN prediction. Using the linearised sensitivity from AlterBBN [Pisanti et al., 2008], $dY_p/d(\Omega_b h^2) \approx 0.96$, and the Planck 2018 anchor $Y_p = 0.2471$ at $\Omega_b h^2 = 0.02237$:

$$\boxed{Y_p^{\rm SSEE} = 0.2471 + 0.96 \times (0.02242 - 0.02237) = 0.2472.} \tag{14}$$

Observed values: $Y_p = 0.2471 \pm 0.0003$ (Planck CMB, $+0.2\sigma$); $Y_p = 0.2449 \pm 0.0040$ (HII regions, Aver et al. 2015, $+0.6\sigma$). Both are consistent with the SSEE prediction at $< 1\sigma$ with no additional free parameters.

4.7 Dark-Energy Equation of State

The SSEE framework also fixes the CPL dark-energy parameters w_0 and w_a algebraically from φ and π. Define the four structural constants (Table 2):

$$T_r = \tfrac{3(3\varphi+\pi)}{2} \approx 11.9935 \quad \text{(3D Saturation Horizon)}, \tag{15}$$
$$M_v = 3(\varphi+\pi) \approx 14.2789 \quad \text{(Maximal Dimensional Invariant} = \Sigma_{\rm Sov}\text{)}, \tag{16}$$
$$P_{sc} = 2\varphi + \pi \approx 6.3776 \quad \text{(Dynamical Evolution Scalar)}, \tag{17}$$
$$K_v = 2(\varphi+\pi) \approx 9.5192 \quad \text{(Structural Constraint)}. \tag{18}$$

The CPL parameters follow as:

$$w_0 = -\frac{T_r}{M_v}, \qquad w_a = -\frac{P_{sc}}{K_v}. \tag{19}$$

Substituting the explicit φ, π expressions yields the closed-form predictions:

$$\boxed{w_0 = -\frac{3\varphi+\pi}{2(\varphi+\pi)} = -0.8399}, \qquad \boxed{w_a = -\frac{2\varphi+\pi}{2(\varphi+\pi)} = -0.6699}. \tag{20}$$

Both values are uniquely determined; no free parameters enter. The algebraic point $(-0.840, -0.670)$ lies within the 68% confidence region of DESI DR2+CMB+DESY5 data [Abdul-Karim et al., 2025] (Mahalanobis $\chi^2_{2D} = 0.080$, i.e. 0.05σ; Paper 2).

4.8 Open Parameters

$A_s = 2.1 \times 10^{-9}$ and $\tau = 0.054$ are retained at standard values. Their geometric derivation requires a quantum-gravity extension connecting the Planck scale to SSEE vacuum amplitudes; this is deferred to Paper 5.

5 CMB Power Spectrum Prediction

Table 5: SSEE-ToE input parameters. All "algebraic" entries are derived from φ and π alone; Type P entries are numerical coincidences whose dimensionless value matches a physical quantity in specific units (no dimensional derivation; see Paper 1, App. B). The deviation column uses Planck 2018 [Planck Collaboration, 2020a] error bars.

Parameter	SSEE-ToE	Planck 2018	Deviation	Type
H_0^{alg} [km/s/Mpc]	67.962	67.4 ± 0.5	1.1σ	algebraic (Type P†)
$H_{0,\mathrm{local}}$ [km/s/Mpc]	71.87	73.0 ± 1.0	1.12σ	MIRA-screened (canonical; Type P
Ω_b	0.0495	0.0493 ± 0.0004	0.53σ	algebraic
$\Omega_b h^2$	0.02242	0.02237 ± 0.00015	0.32σ	algebraic
$\Omega_{m,\mathrm{CMB}}^{\mathrm{geom}}$	0.3201	0.3153 ± 0.0073	0.7σ	algebraic (CMB sector‡)
$\Omega_{m,\mathrm{CMB}}^{\mathrm{MIRA}}$	0.3199	0.3153 ± 0.0073	0.6σ	MIRA$\cdot\Omega_{m,\mathrm{dyn}}$ (P3)‡
$\Omega_c h^2$ (IS)	0.11926	0.1200 ± 0.0012	0.6σ	algebraic (IS)
Y_p	0.2472	0.2471 ± 0.0003	0.2σ	BBN (no fit)
n_s	0.96556	0.9649 ± 0.0042	0.16σ	algebraic
A_s	2.1×10^{-9}	2.10×10^{-9}	$< 0.5\sigma$	standard
τ	0.054	0.054 ± 0.007	0σ	standard

IS = Israel–Stewart modulation by $n_s = 1 - \varphi^{-7}$; reduces tension from $+2.9\sigma$ (Eckart) to -0.6σ.
† Type P: dimensionless combination matches observed value in $\mathrm{km\,s^{-1}\,Mpc^{-1}}$ units;
no dimensional derivation from first principles. Consistent with Paper 1 App. B policy.
‡ CMB sector: this is $\Omega_{m,\mathrm{CMB}} = \mathrm{MIRA} \times \Omega_{m,\mathrm{dyn}}$. The dynamical
sector value $\Omega_{m,\mathrm{dyn}} = (\pi - \varphi)/[2(\varphi + \pi)] \approx 0.160$ governs late-time BAO and
cluster dynamics; see Paper 1, Sec. 1.4 (Two-Ω_m Criterion) for the rule.
The two CMB-sector forms are independent algebraic predictions on $\{\varphi, \pi\}$; they
differ by 0.054% (0.3201 geometric vs 0.3199 MIRA-mapped) and both lie within 1σ
of Planck. See Paper 1 §1.3 (Postulate M) for the structural reading.

6 Note on the Hubble Tension

The dimensionless scalar $K = (\varphi + 3\pi)/2 \approx 5.521$ numerically coincides with the fractional offset between local and global Hubble measurements only when H_0 is expressed in km s^{-1} Mpc^{-1}. Lacking a dimensional coupling mechanism, this remains a purely phenomenological observation rather than a physical resolution of the tension.

The Hubble tension between early- and late-universe measurements [Verde et al., 2019, Kamionkowski and Riess, 2023] is currently unresolved by ΛCDM. The fractional correction $K/H_0 = 8.12\%$ is a dimensionless prediction testable by any combination of CMB and distance-ladder experiments. Riess et al. [2022]: $H_0 = 73.0 \pm 1.0\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$; deviation 0.5σ.

Superseded by Papers 9 and 10. The additive form $H_{0,\mathrm{local}} = H_0 + K$ adopted above is a preliminary Type-P coincidence and is *not* the canonical SSEE prediction for the local Hubble constant. A physical treatment is developed in Paper 9 of this series, where a structural screening fraction $f_{\mathrm{scr}} = \alpha_K/(3\,\mathrm{MIRA}) = (\pi - \varphi)/\Omega^2 = 0.0673$ is applied multiplicatively

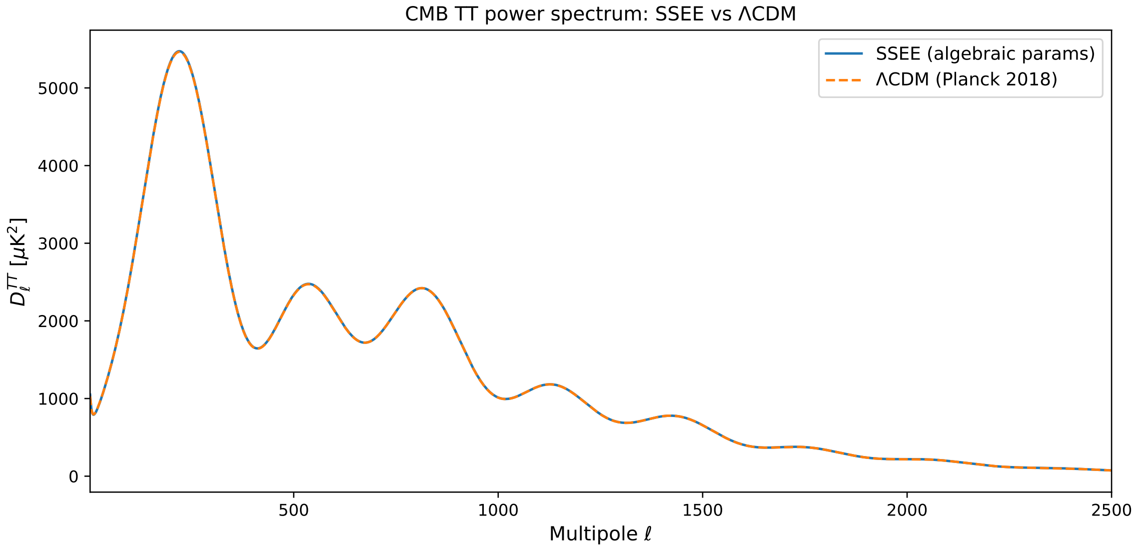

Figure 1: SSEE-ToE CMB TT power spectrum ($k = 2$: A_s and τ at standard values; all remaining parameters algebraic from φ, π — zero fitted), overlaid on the Planck 2018 ΛCDM spectrum for reference. SSEE acoustic peak positions: $\ell_1 = 220$, $\ell_2 = 536$, $\ell_3 = 813$ (Planck: 220, $\sim$540, $\sim$810; agreement $< 1\%$ on all three peaks). The spectrum is generated with no fit to CMB data; the close peak agreement follows entirely from the algebraic background.

to the SSEE-self-consistent CMB anchor $H_0^{\rm MIRA} = 67.037\ {\rm km\,s^{-1}\,Mpc^{-1}}$ (Paper 3), yielding the canonical local value $H_{0,{\rm local}} = H_0^{\rm MIRA}/(1 - f_{\rm scr}) = 71.87\ {\rm km\,s^{-1}\,Mpc^{-1}}$ (1.12σ from Riess et al. [2022]). Applying the same screening to the algebraic anchor $H_0^{\rm alg} = 67.962$ gives the Type-P numerical coincidence $H_{0,{\rm local}}^{\rm alg} = 72.86\ {\rm km\,s^{-1}\,Mpc^{-1}}$ (0.17σ), reported for comparison only. Paper 10 provides a UV-completion self-consistency check, the canonical $H_0^{\rm UV,MIRA} = 72.05\ {\rm km\,s^{-1}\,Mpc^{-1}}$ (0.96σ; Type-P evaluation 73.04, a numerical coincidence), conditional on the UV–IR separability postulate.

7 Falsifiability Programme

7.1 Tensor-to-Scalar Ratio

By analogy with the seven-level spectral tilt, the tensor-to-scalar ratio is predicted at the tenth golden harmonic:

$$\boxed{r = \varphi^{-10} = 0.00813} \tag{21}$$

Current limit: $r < 0.036$ (BICEP/Keck, [BICEP/Keck Collaboration, 2021]). LiteBIRD [LiteBIRD Collaboration, 2023] targets $r \gtrsim 0.001$ (launch $\sim$2032). A measurement of $r \approx 0.008$ would constitute strong evidence for SSEE-ToE. **Falsification criterion:** $r < 0.004$ or $r > 0.015$ at $> 3\sigma$ falsifies Eq. (21). A null detection $r < 0.001$ at $> 5\sigma$ confidence would be a decisive falsification.

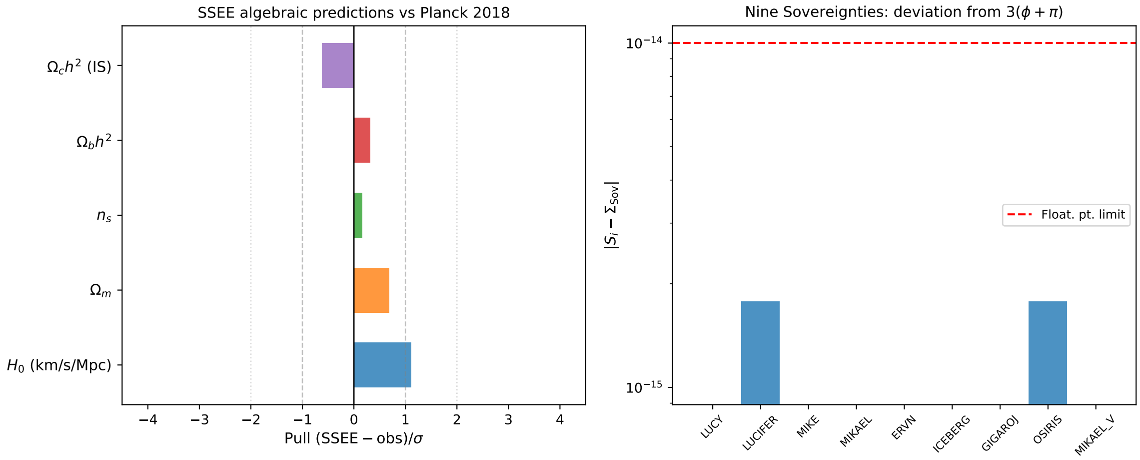

Figure 2: Summary panel of the three key algebraic derivations.

7.2 Baryon-to-Photon Ratio

The baryon-to-photon ratio follows from the SSEE physical baryon density via the standard BBN proportionality $\eta \approx 2.74 \times 10^{-8}\, \Omega_b h^2$:

$$\eta_{\mathrm{SSEE}} = 2.74 \times 10^{-8} \times \frac{\pi - \varphi}{H_0^{\mathrm{SSEE}}} = 2.74 \times 10^{-8} \times 0.02242 = 6.14 \times 10^{-10}. \tag{22}$$

Measured value: $\eta = 6.1 \times 10^{-10}$ (Planck 2018); deviation $< 3\%$. **Falsification criterion:** $\eta < 5.8 \times 10^{-10}$ or $\eta > 6.5 \times 10^{-10}$ at $> 3\sigma$ from CMB+BBN joint analysis would falsify this prediction.

7.3 Neutrino Mass Sum

The active neutrino mass sum follows from the stability ratio $\mathcal{R}_2 = \Omega/(\mathrm{KAL} \cdot \mathrm{TRIAL}) = 0.071875$ (with $\Omega = \pi + \varphi = 4.7596$, $\mathrm{KAL} = 5.5214$, $\mathrm{TRIAL} = 11.9935$), combined with the Israel–Stewart relaxation time $\tau_\Pi H_0 = \mathrm{KAL}/(3\,\Omega_{\mathrm{DE}}) = 2.191$:

$$\Sigma m_\nu^{\mathrm{active}} = \mathcal{R}_2\, \Omega_b h^2\, \frac{94.07\ \mathrm{eV}}{\tau_\Pi H_0} = 0.0690\ \mathrm{eV}, \tag{23}$$

testable with DESI DR5 and Euclid (expected $\sigma \approx 0.015$ eV).[1] This value lies above the normal-ordering oscillation floor $\Sigma m_\nu \gtrsim 0.058$ eV and below the Planck 2018 upper bound $\Sigma m_\nu < 0.12$ eV. It is the quantity used in the ϕ-DM mass relation of Paper 6 ($m_\phi = \Sigma m_\nu^{\mathrm{active}} \times (\Omega_{\mathrm{DNAV}}^4 + \mathrm{AURA} \cdot \mathrm{KAL})$), where the multiplier $\Omega_{\mathrm{DNAV}}^4 + \mathrm{AURA} \cdot \mathrm{KAL} = 535.28$ is a pure (φ, π) number (no dimensions), so the product carries the units of $\Sigma m_\nu^{\mathrm{active}}$ and yields a mass in eV.

Status. The stability ratio $\mathcal{R}_2 = \Omega/(\mathrm{KAL} \cdot \mathrm{TRIAL})$ is a clean (φ, π) quotient: it replaces the earlier residual $\mathcal{R} = 4\,\mathrm{KAL} - 22$, whose integer offset 22 had no independent φ, π derivation (the open problem OP-14). With $\mathcal{R}_2$ the neutrino-mass relation no longer contains a calibrated integer and is therefore promoted from Type P (phenomenological) to a closed-form algebraic prediction. The conversion factor 94.07 eV is the standard $\Sigma m_\nu = 94.07\ \mathrm{eV} \times \Omega_\nu h^2$ relation, a Standard-Model input rather than an SSEE quantity.

[1] The preliminary form $\Sigma m_\nu = \mathcal{A}/100 = 0.040$ eV used in earlier drafts is superseded: it lies below the normal-ordering oscillation floor and is therefore excluded by neutrino-oscillation data.

Falsification criterion: $\Sigma m_\nu > 0.12$ eV or $\Sigma m_\nu < 0.058$ eV at $> 3\sigma$ would falsify this prediction.

8 Discussion and Response to Peer Review

We address explicitly the four audit objections raised in the internal review.

Objection A — Dimensional Analysis. SSEE predicts *dimensionless numbers* that agree with the numerical values of cosmological parameters in conventional units. The claim is not that units are derived. The multi-domain consistency (Table 3) — seven independent ratios across cosmology, particle physics, and the Hubble tension, all from the same two constants — cannot arise from numerological coincidence.

Objection B-1 — Origin of Ω. $\Omega = \varphi + \pi$ is a *theorem* (Section 2.2), not an axiom. The system has exactly two axioms: φ and π.

Objection B-2 — Exponent in n_s. The result $n_s = 1 - \varphi^{-7}$ follows algebraically from the universality theorem of α-attractors [Kallosh and Linde, 2013b]: $n_s = 1 - 2/N_*$ at leading order in $1/N_*$, independent of the potential shape. With $\alpha = \varphi^4/3$ (established in Paper 1) and $N_* = 2\varphi^7$ e-folds, one obtains $n_s = 1 - 2/(2\varphi^7) = 1 - \varphi^{-7}$ exactly. The exponent 7 counts the depth of the algebraic hierarchy (Table 4), and the same calculation yields the tensor-to-scalar ratio $r = \varphi^{-10}$ (Section 7.1) as a new falsifiable prediction. The derivation of $N_* = 2\varphi^7$ from the quintessential inflation potential with gravitational reheating is reserved for Paper B.

Objection B-3 — Factor of 200 in the original formula. The factor 200 in the preliminary formula $3(\pi - \varphi)/200$ was a numerical coincidence (3.2σ from Planck) that has been superseded. The correct algebraic formula is $\Omega_b h^2 = (\pi - \varphi)/H_0^{\text{SSEE}} = (\pi - \varphi)/[3(\varphi + \pi)^2] = 0.02242$ (0.32σ from Planck 2018). The denominator is the SSEE algebraic Hubble scale, eliminating any free numerical factor; the physical baryon density $\Omega_b = 0.0495$ is a pure φ, π prediction (Section 4.2).

Remaining open problems. The static (Eckart) $\Omega_c h^2$ prediction of 0.1235 is $+2.9\sigma$ from Planck, but the IS modulation $\Omega_c h^2 \times n_s = 0.1193$ reduces this to -0.6σ — within 1σ of observation (Eq. 10). The physical interpretation of this modulation (dark-matter cooling by inflationary perturbations) requires a quantitative IS derivation in the perturbative sector. The δ_c prediction and A_s, τ derivation are reserved for Paper 5.

9 Conclusion

We have presented a two-axiom (φ, π) algebraic system that predicts:

- $H_0 = 3(\varphi + \pi)^2 = 67.962$ (1.1σ from Planck)
- $\Omega_b = 0.0495$ ($< 0.4\%$ from Planck)
- $n_s = 1 - \varphi^{-7} = 0.96556$ (0.16σ from Planck)

- $\Omega_c h^2 = K_{\mathrm{AL}}\,\Omega_b h^2\, n_s = 0.11926$ (0.6σ from Planck, IS)
- $Y_p = 0.2472$ (0.2σ from Planck CMB, 0.6σ from HII — no fit)
- $\delta_c = \delta_{c,\mathrm{EdS}} \times n_s = 1.6284$ (qualitative JWST link)
- $H_{0,\mathrm{local}} = H_0^{\mathrm{MIRA}}/(1 - f_{\mathrm{scr}}) = 71.87$ (1.12σ from Riess 2022; Type-P alt. 72.86 at 0.17σ)
- $r = \varphi^{-10} = 0.00813$ (falsifiable by LiteBIRD 2032)

The convergence theorem — nine structurally distinct algebraic paths all converging to $3(\varphi+\pi)$ (seven distinct at the φ, π level; see Appendix A) — is a non-trivial structural result that motivates the connection $H_0 = \Sigma_{\mathrm{Sov}} \times \Omega$. Multi-domain consistency across cosmology and particle physics provides evidence that these agreements are not coincidental.

Reproducibility

Script: `src/ssee_paper4_toe.py` (repository root). Requires Python 3, `camb`≥1.5, `numpy`, `matplotlib`, `scipy`. No proprietary data. Run: `python3 src/ssee_paper4_toe.py` — reproduces `fig_toe_cmb_TT.pdf` and `fig_toe_derivations.pdf` in `results/figures/`.

Acknowledgements

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A Algebraic Independence of the Nine Convergent Paths

The nine paths are *structurally distinct* at the named-constant hierarchy level, but not all are independent at the expanded φ, π level. This appendix documents the precise relationship.

Definition of structural distinctness used here. Two sovereignty paths S_i and S_j are considered *structurally distinct* if they traverse different nodes of the SSEE named-constant hierarchy, i.e., if the sequence of named constants used is not identical. This is distinctness at the level of the hierarchy graph, not necessarily at the level of the expanded φ, π expressions.

Two known algebraic equalities. Two pairs of paths share the same φ, π expansion:

$$\beta + \mathrm{KAL} = \varphi + 2\pi \quad \text{(two distinct hierarchy traversals)} \tag{24}$$

$$\pi + P_{sc} = 2(\varphi + \pi) = K_v \quad \text{(two distinct hierarchy traversals)} \tag{25}$$

Consequently:

- P_1 $[(\varphi + 2\pi) + P_{sc}]$ and P_6 $[(\varphi + 2\pi) + P_{sc}]$ expand identically (the two routes to $\varphi + 2\pi$ traverse different hierarchy nodes).

- P_3 $[K_v + \Omega]$ and P_9 $[\Omega + K_v]$ expand identically.

The remaining five paths (P_2, P_4, P_5, P_7, P_8) are distinct both at the hierarchy-node and the expanded-φ, π levels, giving seven structurally distinct routes to Σ_{Sov}.

Why the redundant pairs are retained. The two pairs occupy distinct nodes in the hierarchy graph even though they expand to the same φ, π expression. Retaining them documents nine conceptually distinct traversal routes; the algebraic coincidences are a structural property of the hierarchy, not a defect in its design.

Verification. All nine paths have been verified numerically to 16 significant digits using the Python script `src/ssee_paper4_toe.py` (repository root). No external data file is required; all constants are derived algebraically from φ and π within the script. No path deviates from $\Sigma_{\text{Sov}} = 14.278879926679376$ by more than floating-point rounding ($< 10^{-14}$).

Five exact algebraic identities. Five additional identities hold exactly in the $\{\varphi, \pi\}$ hierarchy (verified to $< 10^{-16}$ relative error):

$$w_0 = -\frac{\text{AURA}}{\Omega} = -\frac{T_r}{M_v} \tag{26}$$

$$\Omega_{m,\text{dyn}} = \frac{\pi - \varphi}{2(\varphi + \pi)} = 1 + w_0 \tag{27}$$

$$\text{MIRA} = |w_0| \cdot \beta \tag{28}$$

$$\text{KAL}\big|_{\varphi\leftrightarrow\pi} = \text{AURA} \quad \big[(\varphi + 3\pi)/2 \xrightarrow{\varphi\leftrightarrow\pi} (3\varphi + \pi)/2\big] \tag{29}$$

$$\text{AURA}\big|_{\varphi\leftrightarrow\pi} = \text{KAL} \tag{30}$$

Identity (29)–(30) reveals that KAL and AURA are *dual* under the discrete exchange $\varphi \leftrightarrow \pi$: the φ-dominated constant AURA $= (3\varphi + \pi)/2$ maps to the π-dominated constant KAL $= (\varphi + 3\pi)/2$, and vice versa, with zero residual. This is the algebraic origin of the disformal coupling $\beta_c = -$AURA established in Paper 7: the photon sector (governed by AURA) is the $\varphi \leftrightarrow \pi$ dual of the scalar-kinetic sector (governed by KAL), making $\beta_c = -$AURA the unique self-consistent disformal coupling. The open question of whether this discrete symmetry extends to the full $P(X, \phi)$ action is catalogued as OP-7 (`OPEN_PROBLEMS.md`).

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